CardioCog-RAG: the whole project from zero

Read this once, slowly, before presenting. No medical or AI background is assumed. Every number here was checked against the result files in the repository.

1. The project in six sentences

- Some older people lose memory and thinking ability until they cannot manage daily life. That is called **dementia**.
- **Our question:** for a person who does **not** have dementia today, can a computer estimate the chance that they will be diagnosed with dementia **within 3 years**?
- We used a large American research data set (NACC): 57,038 volunteers who visit a clinic about once a year.
- We gave a computer program five kinds of information about each person and let it learn the pattern from people whose outcome we already know.
- It works well (score 0.94 out of 1.0), mostly because it knows how the person's brain is doing **today**.
- The original idea, that heart and blood-vessel health would help predict it, turned out to add very little here. That is a finding, not a failure.

2. The only medical words you need

Word	Plain meaning
Dementia	Memory and thinking loss severe enough that the person cannot manage daily life alone
Alzheimer's disease	The most common cause of dementia
Cognitive	To do with thinking and memory
Cognitively normal	No thinking problem found at the visit
MCI (mild cognitive impairment)	A noticeable thinking problem, but the person still manages daily life. A halfway stage
CDR	A doctor's rating of how much daily life is affected. 0 = no problem. We use the total, 'CDR Sum of Boxes'
Cardiovascular	To do with the heart and blood vessels: blood pressure, diabetes, stroke, heart disease
APOE	A gene. One version of it (called e4) raises the risk of Alzheimer's
Amyloid and tau	Two proteins that build up in the brain in Alzheimer's disease
PET scan	A brain scan that shows how much amyloid or tau is there

3. The only AI words you need

Word	Plain meaning
Model	A program that learns a pattern from examples, then applies it to new cases
Feature	One number the model is given about a person, for example age or a memory score. We use 48
Training	Showing the model people whose outcome we know, so it can learn
Test set	People kept hidden during training, used once at the end as the exam
AUROC	The score. Pick one person who got dementia and one who did not: how often does the model give the higher risk to the right one? 0.5 = coin toss, 1.0 = perfect

Word	Plain meaning
Parameter / setting	A choice we make before training, for example how many decision trees or how many years ahead to predict
Seed	A fixed number that makes the 'random' parts repeat exactly, so results can be reproduced
Leakage	When the model accidentally sees the answer or the future. It makes results look better than they are
SHAP	A method that shows how much each feature pushed one prediction up or down
RAG	A chatbot that first looks up trusted documents and answers only from them

4. The data

Ten files from NACC, kept in `data/raw/`. They are private, so the `data/` folder is never uploaded to GitHub.

File	What is in it	Used?
`investigator_ftldlbd_nacc74.csv`	The main file. One row per clinic visit: 217,598 visits of 57,038 people, 2,649 columns	Yes. All 48 features and the outcome
Two amyloid 'gaain' files	For each amyloid PET scan: the amount of amyloid (Centiloids) and positive / negative	Yes. 2 PET features
Two tau files	For each tau PET scan: the amount of tau in each brain region	Yes. 2 PET features
Two regional amyloid files, FDG file, PET quality list, CLARiTI tracking file	Detailed regional values, another scan type, and bookkeeping	No features taken
`data/raw/pet/`	599 raw brain scan images, about 100 GB	Yes, for the separate image experiment

The five kinds of information (48 features, plus 4 from PET)

Kind	Examples	Features
Demographics	Age, sex, years of education	8
Cardiovascular	Hypertension, diabetes, stroke, blood pressure, smoking, medicines	25
Cognitive	Memory and thinking test scores; the doctor's CDR rating and status	12
Genetics	APOE gene, family history	3
PET	Amount of amyloid and tau	4

Exact column names are in `docs/features_by_category.md`.

5. What is in the project folder

Folder	What it holds
`data/`	The NACC files and everything we compute from them. Private, never on GitHub
`scripts/`	The Python programs. Each does one step. This is where the real work happens
`notebooks/`	12 step-by-step notebooks with explanations and charts. Read these to follow the story
`reports/`	The results the scripts wrote: numbers (.json), summaries (.md) and charts (figures/). Totals only, safe to share
`models/`	The trained model the screen uses. Private, never on GitHub
`app/`	The doctor's screen: `app/web` (React page) and `app/api.py` (runs the model)

Folder	What it holds
`rag/`	The list of research papers and the test questions for the evidence chatbot
`docs/`	Explanations for people: this guide, the simple project note, viva questions, the demo guide

6. The work, step by step

Each step is one script in `scripts/`. 'Settings' are the choices we fixed. Run order is top to bottom.

Step 1. Look at the data before touching it

Script: `audit_data.py`

What it does	Counts rows, columns, people and empty cells in every file. Checks the 599 image folders. Changes nothing.
Result	We learned what is usable: 787 of 2,649 columns have data; 446 people have images.

Step 2. Clean the data, decide who and what, split into groups

Script: `build_dataset.py`

What it does	Turns 'no answer' codes into missing. Compares each test score with healthy people of the same age, sex and education (a z-score). Builds the 48 features. Works out the answer for each person: dementia within 3 years, yes or no. Splits people into training, validation, test and 'unseen hospitals'.
Settings	Prediction window: 3 years. Seed: 2026. 20% of centres (9 of 46) kept aside whole as unseen hospitals. The remaining people split 70% training, 15% validation, 15% test. Only in-person visits. A PET scan counts only if taken up to 365 days before the visit, never after.
Result	19,470 people, of whom 2,818 developed dementia. Training 10,879, validation 2,241, test 2,286, unseen hospitals 4,064.

Step 3. Train five kinds of model and compare them

Script: `train_tabular.py`

What it does	Trains five models on the 48 features, from simple to modern. Then removes one kind of information at a time to see how much each kind matters.
Settings	Logistic regression (the simplest). Random forest: 500 trees. XGBoost: up to 3,000 small trees, stops early when the validation score stops improving, learning rate 0.03. Neural network: two layers of 128 and 64 units. FT-Transformer: 3 layers. Each one except logistic regression is trained 3 times with seeds 0, 1, 2 and averaged.
Result	All five score the same: 0.932 to 0.939 on the test group, about 0.945 at unseen hospitals. Cardiovascular information adds about +0.001.

Step 4. Report separately for normal, MCI and everyone

Script: `train_by_baseline_group.py`

What it does	Scores the same model inside each starting group, and measures how much the doctor's rating drives the score.
Settings	XGBoost, 3 seeds, same groups as step 3.
Result	Everyone 0.937. MCI only 0.835. Cognitively normal only 0.884, but that rests on just 24 cases. Without the doctor's rating, status and thinking tests the score falls to 0.75.

Step 5. Does a person's history help?

Script: `train_longitudinal.py`

What it does	Asks whether looking at earlier visits, not just today's, improves the prediction. Tries simple 'rate of change' numbers and a GRU, a model that reads visits in order.
Settings	People with at least 2 earlier visits. Up to 6 visits read. GRU with 48 memory units.
Result	Only a small gain: +0.000 to +0.005.

Step 6. Explain what the model relies on

Script: `explain.py`

What it does	Uses SHAP to measure how much each feature pushes predictions up or down. Retrains 8 times to check the ranking is stable.

Result	Doctor's rating and status about 41%, thinking tests 35%, demographics 15%, cardiovascular 5%, genes under 5%.

Step 7. Do the PET numbers help?

Script: `train_pet.py`

What it does	Takes the main model's risk and adds the amyloid and tau numbers with a small second model, then checks whether the score improves.
Settings	Two ways of matching a scan to a visit, to be safe.
Result	No clear improvement shown. Few scanned people have been followed long enough.

Step 8. Check the model properly

Script: `validate.py`

What it does	Checks the model in subgroups (sex, age, race, education), checks whether a '40% risk' really means 40%, and repeats the analysis with a different method that includes everyone.
Result	Similar across groups (0.92 to 0.96). Weaker for age 85+. Risks slightly too high at unseen hospitals (15.0% predicted, 13.4% actual).

Step 9. Make a model that copes with missing information

Script: `train_final.py`

What it does	In real life not every patient has every test. During training we hide whole sections at random so the model practises working with gaps ('modality dropout'). Saves this model for the screen.
Settings	Each section hidden with probability 35%, in 4 extra copies of the training data.
Result	With no thinking information at all, a normal model's error is 0.44 (very bad); ours is 0.12.

Step 10. The brain scan images (a separate experiment)

Script: `pet_convert.py, pet_register.py, pet_suvr.py, train_pet_image.py`

What it does	Turns raw scanner files into one 3D picture per scan. Lines all heads up to the same position. Measures amyloid from our pictures and compares with NACC's experts. Trains a 3D CNN (an image model) to say amyloid positive or negative.
Settings	Pictures resized to 80 x 96 x 80. CNN trained for 40 rounds, tested 5 ways so every scan is tested once.
Result	Our measurement matches the experts': 0.98 agreement on 318 scans. The CNN scored 0.96, below the single simple number (0.98). We did not predict dementia from images: only 5 of 446 people with images developed it.

Step 11. The evidence chatbot (the 'RAG' in the name)

Script: `rag_build.py, rag.py, rag_eval.py`

What it does	Builds a small library of research papers. For a question it finds the most relevant paragraphs and answers only from them, showing the source. Does not give personal medical advice. Says so when the library has no answer.
Settings	11 papers (12 were listed; one could not be downloaded), 447 paragraphs. Two searches combined: by meaning and by keywords. Runs fully on the laptop with a small language model.
Result	Found the right paper for 32 of 32 test questions. Flagged all 8 personal-advice questions (it adds a notice and gives general information only) and declined all 6 off-topic questions. The small language model wrote a proper answer only about 1 time in 5; otherwise the chatbot quotes the papers directly.

Step 12. Check our own work

Script: `verify_results.py, check_clinical_behaviour.py`

What it does	17 checks that the results are honest: nobody in both training and test, no information from the future, answers recomputed a second way, shuffled answers give a coin-toss score. 28 more checks that the screen behaves sensibly.

Result	17 of 17 passed. They earlier caught two real mistakes, which we fixed. 27 of 28 passed; the one failure is reported openly (risk about 8 points too high for people under 65 with MCI).

Step 13. The doctor's screen

Script: `inference.py, app/api.py, app/web`

What it does	A web page. The doctor fills in what is available; the page shows the 3-year risk, what happened to similar people, and what pushed the risk up or down. It warns when entries contradict each other.
Result	A working research prototype. It does not diagnose and does not suggest treatment.

7. The 12 notebooks

Notebook	What it shows
01 to 04	Getting to know the data: the files, visits over time, the kinds of information, the PET data
05	Cleaning, who is studied, what is predicted, the fair split
06	The five models, which information matters, and whether history helps
07	SHAP: what the model relies on
08	Do PET numbers help?
09	Checking the model properly
10	Coping with missing information
11	The brain scan images
12	The evidence chatbot

8. The results to remember

Question	Answer
How many people and visits?	57,038 people, 217,598 visits
Who was studied?	19,470 people without dementia at their first visit; 2,818 developed dementia within 3 years
How good is the model?	0.94 for everyone together; 0.84 among people with MCI; 0.88 among normal people (few cases)
At hospitals it never saw?	About 0.95 at 9 unseen centres
Is fancy AI better?	No. All five models score the same
Does cardiovascular information help?	Little extra value in this group and setup: about +0.001
Does history help?	Only slightly: +0.000 to +0.005
Do PET numbers help?	Not shown
What drives the prediction?	How the brain is doing today: the doctor's rating, status and thinking tests
Is it trustworthy?	17 of 17 honesty checks pass; two mistakes were found and fixed

9. What to say tomorrow

A two-minute version

"Our project asks: if a person does not have dementia today, can we estimate their chance of being diagnosed within 3 years? We used NACC, a large American research data set with 57,000 volunteers. For each person we used five kinds of information: demographics, heart and blood-vessel health, thinking tests with the doctor's rating, the APOE gene, and brain scan numbers. We studied 19,470 people and tested the model on people and on whole hospitals it had never seen. It scores 0.94, where 0.5 is a coin toss and 1.0 is perfect. But we are careful with that number: most of it comes from how the person's thinking is today. Among people who already have mild impairment the score is 0.84. Our starting idea was that cardiovascular information would help. In this data it added very little once thinking tests were known. We report that as a finding. Simple and advanced models performed the same. We also processed 599 raw brain scans and confirmed our measurements match the experts'. Finally we built a screen for a doctor and a chatbot that answers only from research papers. It is a research prototype. It does not diagnose and does not recommend treatment."

A slide order that works

- The question (one sentence) and why it matters
- The data: 57,038 people, five kinds of information
- Who we studied and the fair test (training, test, 9 unseen hospitals)
- Results: the five models and the 0.94 / 0.84 / 0.88 table
- What matters: thinking today, not cardiovascular or history
- PET numbers and the image experiment
- How we checked ourselves: 17 checks, two mistakes found and fixed
- The screen (live demo) and the chatbot
- Limits, and what the project does not claim

The live demo (on the project laptop, in the project folder)

```
python -m uvicorn app.api:app --port 8000
# then open http://localhost:8000 in a browser

python scripts/verify_results.py     # shows 17 of 17 checks passed (a few minutes)
```

Start the screen before the presentation, not during it. Step-by-step demo actions are in `docs/demo_and_test_guide.pdf`.

10. Questions you will probably get

Question	Short answer
Is 0.94 too good to be true?	It mixes two very different groups. People with MCI convert 41.5% of the time, normal people 1.6%. Just knowing the group already gives 0.84. Inside MCI the model scores 0.84, which is the fairer number.
Isn't using the doctor's rating cheating?	No. It is recorded at the first visit and the outcome comes later. But it does mean the model mostly measures how far along someone already is, and we say so.
Why did cardiovascular not help?	People join at about age 71. Blood-vessel damage happens in middle age, which we cannot see, and its effect is probably already showing in the thinking tests. We do not claim heart health is unimportant.
How do you know the model did not see the answers?	Nobody is in both training and test. 9 whole hospitals were never seen. With shuffled answers the score drops to 0.51. Features were checked to use nothing from later visits.

Question	Short answer
Why not a deep model on the scans?	Only 5 of the 446 people with images developed dementia. Nothing can learn from 5 examples, so we did not pretend.
What did the checks find?	Two cases of using the future: education filled in from a later visit, and scans dated after the visit. Both fixed; results changed only in the third decimal.
Why is it called CardioCog-RAG?	Cardio: the cardiovascular question. Cog: cognitive data. RAG: the evidence chatbot design.
Can doctors use it?	No. It is a research prototype tested on past research data only.

11. What not to say

- Do not say the model **diagnoses** dementia. It estimates a risk.
- Do not say cardiovascular health is **useless** or unimportant. Say it added little **in this group and setup**.
- Do not say PET does not help. Say we **could not show** that it helps.
- Do not quote 0.94 alone. Give 0.84 for MCI with it.
- Do not say it was tested on patients. It was tested on past research volunteers, who are mostly White and highly educated.
- If you do not know an answer, say "I will check that in our results" and point to `docs/claims_verification.md`.

If you have one more hour: read `docs/project_explained_simply.md` (the longer version of this guide) and `docs/viva_notes.md` (more questions with answers).